

★ Mixtures and Alligation.

Puzzle:-

$A \Rightarrow 40$; $B \Rightarrow 1$; $C = 0.2$.

Total Money: 280

Total no. of product: 280.

$$A + B + C = 280$$

$$A + B + \frac{1}{5}C = 280$$

$$A + C = 40A + \frac{1}{5}C$$

$$39A = 0.8C$$

$$\frac{A}{C} = \frac{0.8}{39} = \frac{8}{390} \Rightarrow \frac{4}{195}$$

$$A \Rightarrow 4 \Rightarrow 4 \times 40 \Rightarrow 160$$

$$C \Rightarrow 0.2 \Rightarrow 0.2 \times 195 \Rightarrow 39$$

$$A + C = 199 \quad | \quad 199$$

$$B = 280 - 199$$

$$B = 81$$

Mixtures and Alligation.
 $\Rightarrow$ Based on % profit & loss

₹ 128

$$A \Rightarrow 160$$

$$C \Rightarrow 39$$

$$B \Rightarrow 81$$

No

$$A \Rightarrow 4$$

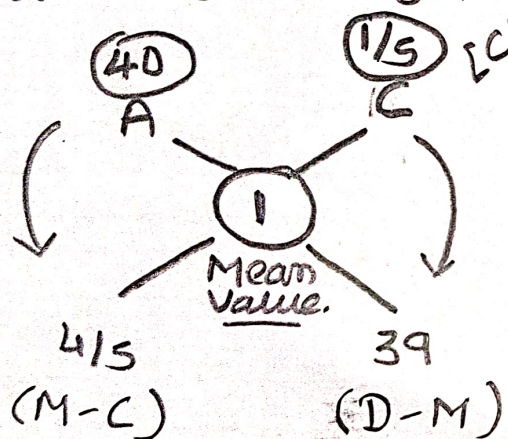
$$C \Rightarrow 195$$

$$B \Rightarrow 81$$

[Dearest Value]

OR.

[cheapest value]



$$A : C = \frac{4}{5} : 39$$

$$A : C = 4 : 195$$

$\therefore$ Mean Value > cheapest value

$\therefore$ Mean Value < Dearest Value.

Que

A trader purchases two varieties of Rice 'A' & 'B' at the rate of ₹48/kg and ₹60/kg respectively. In what ratio he should mix the two varieties, so that the mixtures cost becomes ₹52/kg?

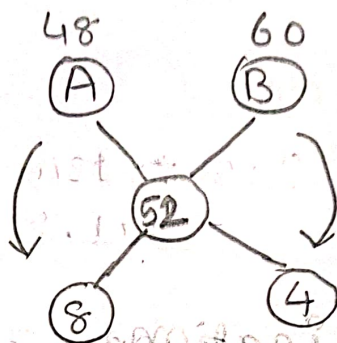
Sol

$$A = 48 \text{ kg}$$

$$B = 60 \text{ kg}$$

$$\Rightarrow A:B = 8:2$$

$$\Rightarrow A:B = 2:1$$



Que

A trader purchases two varieties of Sugar 'P' & 'Q' at the rate of ₹42/kg and ₹80/kg respectively. In what ratio he should mix the two varieties, so that he can sell the mixture at ₹78/kg by getting 30% profit.

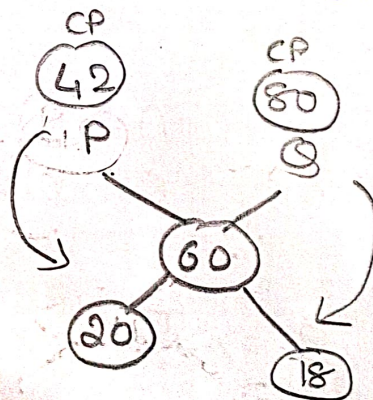
Sol

$$78 \Rightarrow \text{S.P}$$

$$\frac{\text{SP}}{\text{CP}} = 1.3 \quad (30\% \text{ profit})$$

$$\frac{78}{\text{CP}} = 1.3$$

$$\text{CP} = 60$$



$$P:Q \Rightarrow 20:18$$

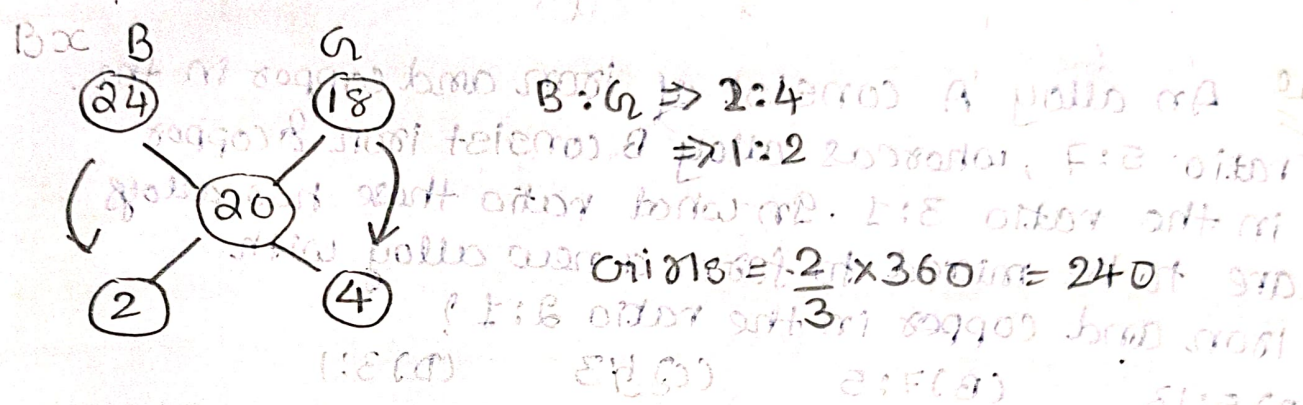
$$P:Q \Rightarrow 10:9$$

Ques

In a class with 360 students, the average age of all the boys is 24 years whereas average age of all the girls is 18 years. If average age of whole class is 20 years, then find the no. of girls in that class.

- (A) 300
- (B) 40
- (C) 120
- ☒ (D) 240

Sol

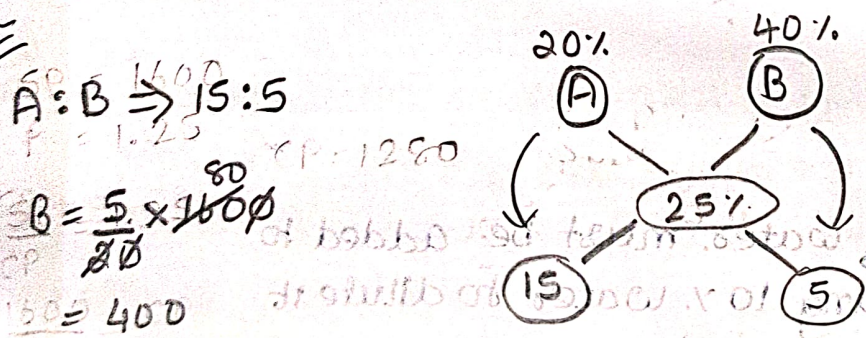


Ques

A merchant sells 1600 horses at an overall profit of 25%. Some of them he sold at 20% profit whereas remaining at 40% profit. Find the no. of horses he sold at 40 profit.

- ☒ (A) 400
- (B) 1200
- (C) 1240
- (D) 300.

Sol



Ques

In a mixture of 40 litres, the ratio of milk and water is 4:1. How much water must be added to this mixture so that the ratio of milk and water becomes 2:3?

- (A) 20 litres
- (B) 30 litres
- (C) 32 litres
- ☒ (D) 40 litres

Sol
Normal method

Mixture = 40L (4:1) adding water

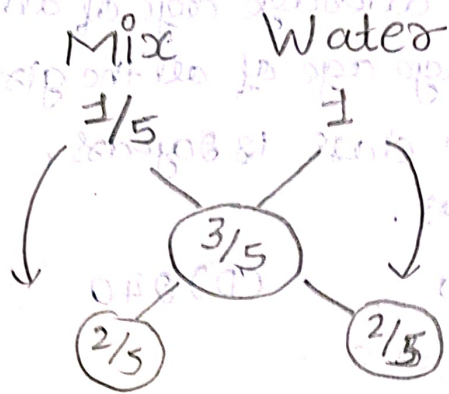
Milk	Water
32L	8L

(I) $\therefore \frac{8+x}{32} = \frac{3}{2}$
 $\Rightarrow 16+2x = 96$
 $\Rightarrow x = 40$

(II) $\frac{8+x}{40+x} = \frac{3}{5}$
 $\Rightarrow 40+5x = 120+3x$
 $\Rightarrow 2x = 80$
 $\Rightarrow x = 40$

(III) $\frac{32+x}{40+x} = \frac{2}{3}$
 $\Rightarrow 96 = 80 + 2x$
 $\Rightarrow 16 = 2x$
 $\Rightarrow x = 8$

① Mixture Alligation Method



$$\text{Mix} : \text{Water} = \frac{2}{5} : \frac{2}{5}$$

$$\text{Mix} : \text{Water} = 1 : 1$$

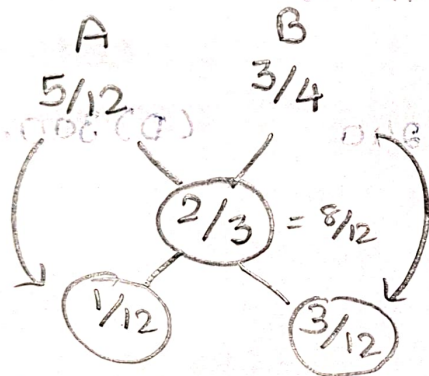
Opt. D

Que An alloy A consist of iron and copper in the ratio 5:7, whereas alloy B consist iron & copper in the ratio 3:1. In what ratio these two alloys are to be mixed to form a new alloy with iron and copper in the ratio 2:1?

- (A) 5:13 (B) 7:5 (C) 1:3 (D) 3:1

Sol

IRON



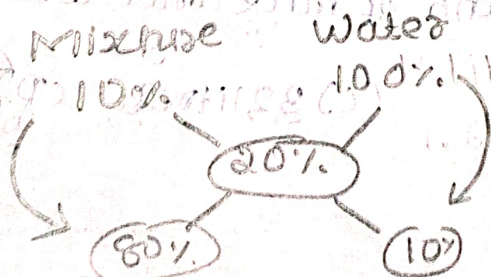
$$A : B = \frac{1}{12} : \frac{3}{12}$$

$$A : B = 1 : 3$$

Que How many litres of water must be added to 16 litres milk containing 10% water to dilute it with 20% water?

- (A) 3 litres (B) 2 litres (C) 1 litre (D) 4 litres

Sol



$$M : W = 8 : 1$$

$\therefore$ 16 litre mixture
 $\Rightarrow$ 2 litre water

Basics

[Assignment]

Que

A mixture contain milk and water in the ratio 5:1. On adding 5 litres of water, the ratio of milk to water becomes 5:2. The quantity of milk in the mixture earlier was.

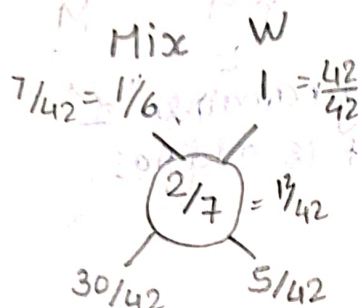
(A) 16 litres

(B) 25 litres

(C) 32.5 L

(D) 22.75 L

Sol



$\frac{5}{6} \times 30 = 25$ litres milk

> litre

Mix: Water

(30):5

6:1

Que

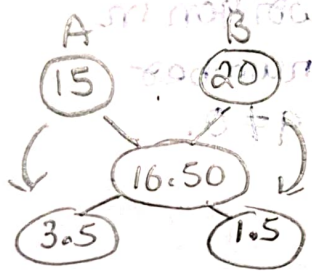
In what ratio must a grocer mix two varieties of pulses costing ₹15 and ₹20 per kg respectively so as to get a mixture worth ₹16.50 per kg?

(A) 3:8

(B) 5:3

(C) 7:3

(D) 4:5



Que

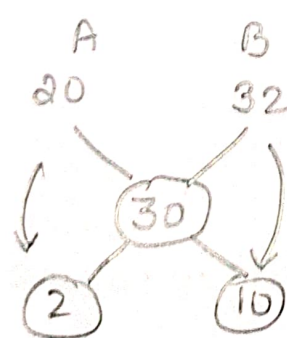
In what ratio must Ankit mix two varieties of sugar worth ₹20 per kg and ₹32 per kg so that by selling the mixture at ₹36 per kg he may gain 20%?

(A) 1:5

(B) 3:5

(C) 2:3

(D) 3:2



SP = 36
 n = 1.2
 CP = (?)
 = 30

$\frac{SP}{CP} = n$

$\frac{36}{CP} = 1.2$

CP = $\frac{3.6}{1.2} \times 30$